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ORIGINAL PAPER

Elastic Wave Modulation of Metamaterial Beams in Bidirectional Asymmetric Acoustic Black Holes: Finite Element Method and Experiment

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Abstract

Purpose Acoustic black hole (ABH) is a passive vibration damping technique that can efficiently mitigate vibrations in beam structures. However, the bandgap performance in most conventional periodic ABH structures is constrained to the flexural direction, and the unit cell possesses a single symmetric configuration. Therefore, this paper proposes a bidirectional asymmetric ABH metamaterial beam to investigate its bandgap characteristics in both the flexural and longitudinal directions.

Method The finite element method was employed to predict the bandgap structure of the bidirectional asymmetric ABH metamaterial beam in various directions, along with the attenuation of evanescent waves within the bandgaps, which was subsequently validated by experimental results. Additionally, the influence of geometric parameters and material damping on the bandgap performance of this metamaterial beam was also examined.

Results The results suggest that, compared to conventional unidirectional metamaterial beams, this metamaterial beam exhibits a wider unidirectional bandgap, several comparatively broader complete bandgaps, and a higher evanescent wave attenuation factor. The formation of flexural bandgaps can be attributed to the combined effects of the ABH effect and Bragg scattering, whereas the formation of longitudinal bandgaps results solely from the Bragg scattering effect. It was further observed that the formation of complete bandgaps arises from the coupling and decoupling of these two effects. Parametric analysis demonstrates that tunable bandgaps in various directions can be realized through uniform changes in geometric parameters. Material damping has a negligible effect on bandwidth but significantly improves the attenuation of flexural waves, while having only a minor influence on the attenuation of longitudinal waves.

Conclusion Numerical analysis and experimental results demonstrate the superior vibration damping performance of this metamaterial beam in both the flexural and longitudinal directions. The proposed structure could provide a novel perspective on the design of multi-directional vibration beams.

Keywords Bidirectional asymmetric ABH metamaterial beams · Bandgap · Flexural wave · Longitudinal wave

Introduction

Metamaterials are artificially engineered to possess physical properties unattainable by natural materials. Based on different disciplines, metamaterials can be categorized into electromagnetic, optical, and mechanical types [1–3]. Electromagnetic and optical metamaterials, exhibiting characteristics such as negative magnetic permeability, negative refractive index, and negative dielectric constant, can be applied in invisibility cloaks, antennas, and various types of sensors [4–6]. Mechanical metamaterials, including acoustic and elastic types, utilize properties such as negative mass, negative modulus, and negative Poisson's ratio, and can be

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used for sound focusing, sound wave attenuation, and energy absorption [7–9]. In recent years, the application of metamaterials in vibration attenuation has garnered widespread attention. Relying on their unique structures and properties, vibration-attenuating metamaterials have shown tremendous potential in various fields, including aerospace, automotive manufacturing, civil engineering, and machinery [10–14]. The mechanism for vibration and noise reduction in metamaterials is attributed to bandgaps in the periodic structural dispersion curve, which prevents the propagation of elastic waves within the bandgap frequency range [15].

To fulfill the significant potential in vibrations, diverse metamaterial structures have been designed to broaden the bandgap. Studies have demonstrated that bandgaps arise through two primary mechanisms. One is the local resonance effect due to localized oscillatory modes [16], and the other is the Bragg scattering effect from wave interference phenomena in periodic lattice structures [17]. The former was first proposed by Prof. Liu [18], who embedded resonant units, such as mass blocks and springs, into the matrix material to form an m–k-c system. This system resonates at specific frequencies, absorbing and dissipating vibrational energy [19–22]. This structure is distinguished by a lattice constant significantly smaller than the wavelength corresponding to the bandgap frequency, allowing the control of large wavelengths with small dimensions. The latter forms a bandgap at a specific frequency by designing the material into a periodic structure [23–26], which requires the lattice size to be of the same order of magnitude as the wavelength of the target bandgap, resulting in huge structural dimensions, particularly for low-frequency vibration. Therefore, effectively enhancing the coupling effect between these two bandgaps is crucial for achieving a broad frequency bandgap. He et al. [27] and Wan et al. [28] proposed localized resonators with composite metamaterial columns and metamaterial cylinders with a tower lattice, respectively, to hinder elastic wave propagation. Wang et al. [29] devised a multi-resonator structure mounted on a beam to provide low-frequency wideband vibration attenuation. Dong et al. [30] proposed a dual-mass membrane system as a resonator for metamaterial plates, which can significantly suppress the vibration bandwidth. Ren et al. [31] designed a bidirectional re-entrant metamaterial aimed at attenuating low-frequency vibrations. Li et al. [32] developed a graded labyrinth of metamaterials capable of generating numerous useful low-frequency bandgaps. Wang et al. [33] proposed a particle-damped resonator to expand the available bandwidth. Hu et al. [34] investigated how graded localized resonators reduce vibration within broad bandgaps. Additionally, various resonators have been designed, including sandwich [35–39], thin plate [40–44], and spiral types [45–48].

Although metamaterials embedded with various types of localized resonators have the potential to enhance bandgap performance, the addition of resonators or mass-spring systems increases the mass of the structure, and their fabrication process is complex and requires sophisticated structural design. A novel elastic wave-absorbing structure called the acoustic black hole has been proposed to address this issue. Acoustic black holes are wedge-shaped structures whose thickness decreases according to a specified power-law relationship $h(x) = \varepsilon x^m (m \geq 2)$. When a flexural wave is transmitted from an equal-thickness zone to an acoustic black hole region, the local phase and group velocities decrease with thickness; the wavelength shortens, and the amplitude increases. Under ideal conditions, a zero-thickness point is reached when the wave velocity drops to zero, the cumulative phase reaches infinity, zero reflection occurs, and the fluctuating energy accumulates at the tip [49–53]. Applying ABH structures as crystalline cells in metamaterials has recently sparked widespread interest due to their apparent advantages in localized energy concentration. Gao et al. [54] analyzed the low-bandgap characteristics of ABH in V-folded metamaterial beams. Lyu et al. [55] proposed a metamaterial plate with an ABH structure embedded in a hexagonal lattice wall, which was found to have an extremely wide bandgap and significant attenuation. Li et al. [56] developed a spiral ABH resonator without compromising its main strength and stiffness. It provided substantial vibration reduction in the low-frequency region using finite-element and experimental methods. Gao et al. [57] used a gradient-arranged rainbow metamaterial beam to obtain a broader bandgap and a more substantial damping effect. Bao et al. [58] proposed a thickness, density, and modulus tri-gradient ABH metamaterial beam based on gradient grading of a single ABH structure that exhibits excellent wave attenuation in the bandgap region. Deng et al. [59] and Zhang et al. [60] examined embedded and graded bilobed ABHs with subwavelength bandgaps, respectively.

As previously stated, most existing studies provide conclusive evidence that metamaterial structures composed of ABH lattices are uniquely suited to opening the flexural wave bandgap. However, only some studies have proposed methods to modulate flexural and longitudinal wave bandgaps. Gao et al. [61] discovered elastic wave attenuation in hollow metamaterial beams embedded with ABH structures, forming bandgaps in various directions. However, the physical mechanism triggering the development of longitudinal bandgaps was not thoroughly explored. In ABH metamaterial plates, the frequency range of the complete bandgap has been identified [62, 63], but the bandwidth is relatively narrow. Bao et al. [64] proposed a helical metamaterial conical rod to address this issue, identifying its dynamic characteristics that couple longitudinal and bending vibration modes. This characteristic broadens the effective frequency range of the complete bandgap and enhances wave attenuation capability. However, the manufacturing process of this design

is complex. Therefore, a metamaterial component that can simultaneously control the propagation of elastic waves in both longitudinal and flexural directions is urgently needed while maintaining a simple structural design.

To our knowledge, the characteristics of hierarchical metamaterials lie in their material properties or geometric configurations that exhibit spatial gradient variations, significantly broadening bandgaps and resulting in a rainbow phenomenon [65–67]. Furthermore, the attenuation ability of acoustic black holes in flexural waves is undeniable. Inspired by the hierarchical structure, a pair of ABHs were flipped by 180° to form a bidirectional asymmetric ABH structure exhibiting spatial gradient-like variations, thereby promoting wave interference phenomena.

Therefore, this paper proposes a bidirectional asymmetric ABH metamaterial beam and investigates its dynamic characteristics in flexural and longitudinal directions. Numerical analysis based on the finite element method, including the ω(k) and k(ω) methods, is employed to predict the complex band structure and the attenuation of evanescent waves in the bandgap. The results indicate the presence of relatively wide directional bandgaps in the dispersion curves, particularly for longitudinal waves. Additionally, several complete bandgaps are observed. The influence of geometric parameters on directional bandgaps is examined, elucidating the physical mechanisms behind their formation. Experimental results confirm the vibration reduction performance of the designed finite periodic structures.

The paper is organized as follows: Sect. "Geometric modeling of bidirectional asymmetric ABH metamaterial beam" details the design model of the bidirectional asymmetric ABH metamaterial beam. Sect. "Complex band analysis of bidirectional asymmetric ABH metamaterial beam" introduces the real and complex band analysis methods within finite element modeling, analyzing numerical results and parametric studies, including band structure, bandgap mechanisms, and the effects of geometrical parameters and material damping. Meanwhile, the attenuation performance of different configurations was characterized using the imaginary wave vector, and their vibration-damping capabilities were compared by analyzing the results of the vibration transmission curves. Sect. "Experimental study of bidirectional asymmetric ABH metamaterial beam" experimentally verifies the attenuation capacities for flexural and longitudinal waves, with the final section presenting the conclusions.

Geometric Modeling of Bidirectional Asymmetric ABH Metamaterial Beam

This section focuses on introducing the geometric design of the bidirectional asymmetric ABH metamaterial beam. The unit cell consists of a pair of ABH structures rotated by 180°, which are subsequently arranged in a linear array along the x-axis. Figure 1 provides a schematic illustration of the thickness profile of the bidirectional asymmetric ABH metamaterial beam. To begin, the uniform and power-law thickness profiles are represented as follows:

$$h(x)=\begin{cases} h_2 & -(l_1+l_2)<x<-l_1 \\ \frac{h_2-h_1}{l_1^m}|x|^m+h_1 & -l_1<x<l_1 \\ h_2 & l_1<x<l_1+2l_2 \\ -\frac{h_2-h_1}{l_1^m}|x-s|^m+t & l_1+2l_2<x<3l_1+2l_2 \\ h_2 & 3l_1+2l_2<x<3l_1+3l_2 \end{cases} \tag{1}$$

The horizontal distance from the origin is denoted by x. The uniform and residual thicknesses of the ABH cell tube in Region I are indicated by $h_2=8$ mm and $h_1=1$ mm, respectively. The residual thickness of the ABH cell tube in Region II is marked by $t=1$ mm. The lengths of the conical and uniform areas of the ABH tube in Region I are identified by $l_1=50$ mm and $l_2=5$ mm, respectively, and the distance between the centers is described by $s=110$ mm. For clarity, Region I is designated the inner concave region, and Region II is the outer convex region. The power index of the conical region is $m=2$, and the position of the rotary shaft is $r=8$ mm. Considering the cost of structural design and ease of later-stage processing, the photosensitive resin was selected as the linear elastic material, with a density

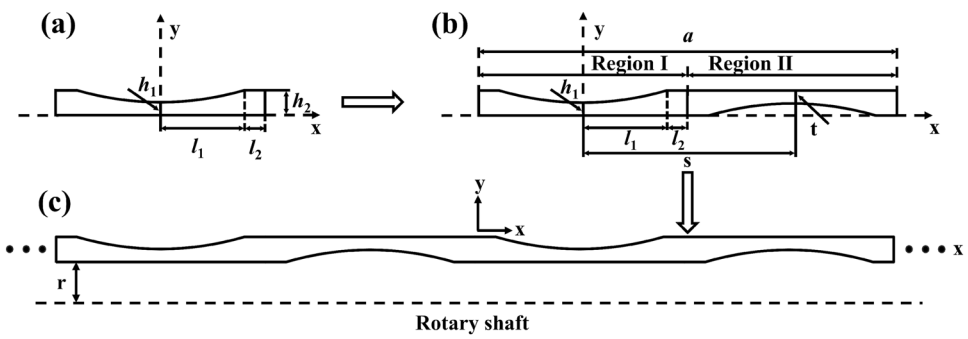


Fig. 1 **a** Thickness profile of a pair of ABH beams; **b** thickness profile of a bidirectional asymmetric ABH beam cell; **c** thickness profile of a bidirectional asymmetric ABH metamaterial beam

of 1160 kg/m^3, elastic modulus of 2.46 GPA, and Poisson ratio of 0.23. Subsequently, it was designed as a bidirectional asymmetric ABH metamaterial beam by connecting two identical profiles at the head and tail.

Complex Band Analysis of Bidirectional Asymmetric ABH Metamaterial Beam

$\omega(k)$ and $k(\omega)$ methods in band analyses

The bidirectional asymmetric ABH metamaterial beam designed in this study is shown in Fig. 1(c). The axial direction of the beam features a one-dimensional acoustic black hole structure composed of periodic subunits, with the x direction coinciding with the axial direction as per the Cartesian coordinate system right-hand rule. To explore the band performance of metamaterials, the elastic wave dynamics equations in each isotropic, homogeneous three-dimensional solid phononic crystal structure are represented as follows [68]:

$$\rho(r)\ddot{u} = (\lambda(r) + \mu(r))\nabla(\nabla \cdot u) + \mu(r)\nabla^2 u \tag{2}$$

where ρ is the density,λ is the first Ramey constant,μ is the second Ramey constant,$u = (u_x, u_y, u_z)$ is the displacement field vector, $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is the Laplace operator.

The displacement field in a one-dimensional phononic crystal, with $\boldsymbol{r}(x, t)$ representing the position vector, is independent of y and z. Therefore, all terms in (2) involving partial derivatives concerning y and z are zero. The wave equation for a one-dimensional phononic crystal in the Cartesian coordinate system is thus obtained as:

$$\begin{cases} \rho(x)\frac{\partial^2 u_x}{\partial t^2} = \frac{\partial}{\partial x}\left[\lambda(x)\frac{\partial u_x}{\partial x}\right] + \frac{\partial}{\partial x}\left[2\mu(x)\frac{\partial u_x}{\partial x}\right] \\ \rho(x)\frac{\partial^2 u_y}{\partial t^2} = \frac{\partial}{\partial x}\left[\mu(x)\frac{\partial u_y}{\partial x}\right] \\ \rho(x)\frac{\partial^2 u_z}{\partial t^2} = \frac{\partial}{\partial x}\left[\mu(x)\frac{\partial u_z}{\partial x}\right] \end{cases} \tag{3}$$

According to Bloch theorem, the displacement field is expressed as follows:

$$u(x, t) = u_k(x)\exp(i(kx - wt)) \tag{4}$$

where k is the Bloch wave vector confined to the first Brillouin zone, w is the angular frequency of the wave, and $u_k(x)$ is a periodic function corresponding to the periodic structure, which can be expressed as follows:

$$u_k(x) = u_k(x + R_n) \tag{5}$$

where R_n is the positive lattice vector. By substituting Eq. (5) into Eq. (4), the periodic boundary conditions can be written as follows:

$$u(x + R_n) = u(x)\exp(ik \cdot R_n) \tag{6}$$

The function $u(x + R_n)$, resulting from translating the displacement field function u_x in a periodic field by an arbitrary positive lattice vector, has an additional phase factor $\exp(ik \cdot R_n)$ of modulus one compared to $u(x)$. By substituting Eq. (6) into Eq. (3), the eigenvalue equation is obtained:

$$(\boldsymbol{K} - w^2\boldsymbol{M})\boldsymbol{U} = \boldsymbol{0} \tag{7}$$

where $\boldsymbol{K}$ is the stiffness matrix, $\boldsymbol{M}$ is the mass matrix, and $\boldsymbol{U}$ is the displacement vector for all nodes. The dispersion relation $\omega(k)$ for periodic structures can be determined by scanning the first Brillouin wave vector k. Due to the symmetry of the first Brillouin zone, it is only necessary to scan the wave vector k along the boundary of the first minimal Brillouin zone $[0,\ \pi/a]$ to obtain the eigenfrequency and plot the real band dispersion curves.

The imaginary part of the band structure cannot be derived using the $\omega(k)$ analysis since only the real wave number can be calculated. Consequently, a method for calculating complex band dispersion curves is proposed based on the COMSOL Multiphysics 6.1 partial differential equation (PDE) module using coefficients. At a specific circular frequency w, an eigenvalue wave vector with both real and imaginary parts is generated, with the imaginary portion reflecting the attenuation capabilities of the evanescent waves in the bandgap between the real parts. Appendix A provides an explicit description of the $k(\omega)$ method.

The Solid Mechanics Module in COMSOL was utilized to compute the band structure of a three-dimensional single cell of a bidirectional asymmetric acoustic black hole metamaterial to obtain the real band dispersion curves. Bloch-Floquet periodic boundary conditions were imposed along the x direction on the two end faces of the cell. To ensure convergence of the results within 10.5 kHz, the model was discretized using quadratic Lagrange elements with at least 10 elements per local wavelength, resulting in a mesh count of 59,497. The minimum element size was 0.05 mm.

To study the propagation patterns of flexural and longitudinal waves in the structure, this paper uses the ratio of the absolute value of each displacement field component to the sum of the absolute values of the displacement field components to express the polarization β in the three directions x, y, and z. The specific expressions are given below:

$$\begin{aligned} \beta_x &= \frac{abs(u)}{abs(u)+abs(v)+abs(w)} \\ \beta_y &= \frac{abs(v)}{abs(u)+abs(v)+abs(w)} \\ \beta_z &= \frac{abs(w)}{abs(u)+abs(v)+abs(w)} \end{aligned} \tag{8}$$

where u, v, w are the components of the displacement field vectors,β_x is the in-plane polarization, and β_y, β_z is the

out-of-plane polarization, and the longitudinal and flexural modes are polarized when the value of $\beta_j (j = x, y, z)$ tends to one.

Band Structure and Mechanisms

Figure 2(a) and (c) illustrate that the dispersion curves of the real part of the complex band of the PDE align closely with the real band dispersion curves of the FEM. This confirms the accuracy of the complex band analysis and demonstrates the interoperability of the $\omega(k)$ and $k(\omega)$ methods. There are six flexural wave bandgaps, labeled BG1, BG2, BG3, BG4, BG5, and BG6, within the range of 8000 Hz. The frequency ranges of these bandgaps are 224.98–364.87 Hz, 1063.6–1405.1 Hz, 2401.3–2726.4 Hz, 3616.9–4970.2 Hz, 5995–6157.3 Hz, and 7585.9–7680.6 Hz. To further reflect the evanescent wave attenuation ability at the bandgap, the imaginary part of the complex band is plotted, as shown in Fig. 2(b), illustrating that the evanescent wave diminution ability is greatest in BG4, consistent with the dispersion curves of the infinite-period metamaterial beams. Meanwhile, the subsequent analysis of the longitudinal bandgap reveals that in the frequency range of BG4, 3616.9–3730.8 Hz constitutes a unidirectional bandgap, and 3730.8–4970.2 Hz constitutes a complete bandgap. This transition not only transforms the unidirectional bandgap into a complete bandgap but also significantly enhances vibration-damping capability.

To better visualize the bandgap characteristics of metamaterial beams against longitudinal vibration, 2D axisymmetric modeling was conducted to eliminate the effects of torsional and local extrusion modes on the longitudinal

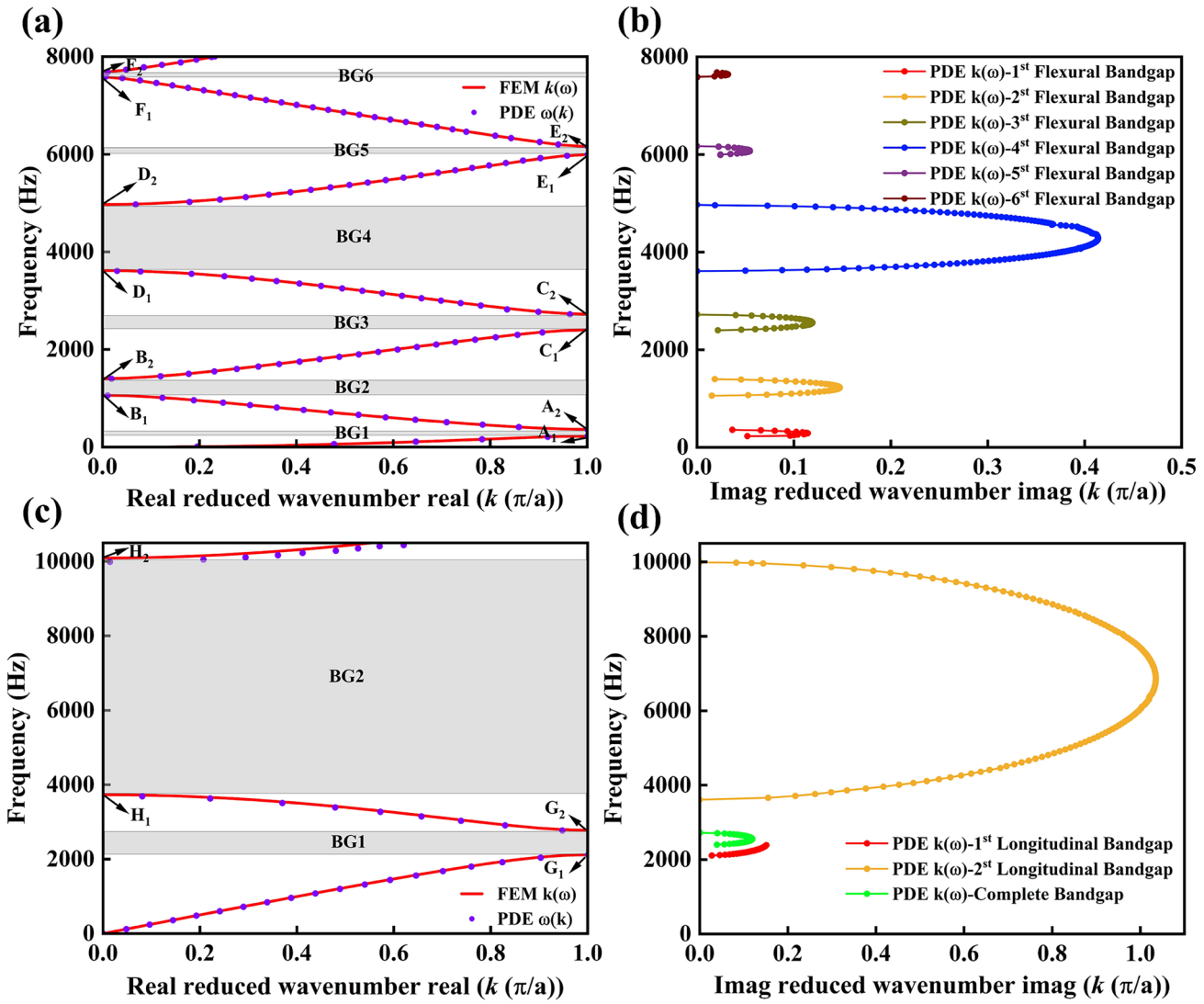


Fig. 2 **a** and **c** Real band flexural and longitudinal dispersion curves of bidirectional asymmetric ABH metamaterial beam, respectively; **b** and **d** flexural and longitudinal imaginary part dispersion curves of bidirectional asymmetric ABH metamaterial beam, respectively

modes introduced by the 3D model. Figure 2(c) indicates that the longitudinal bandgap eigenfrequency increases to 10.5 kHz as the upper boundary of the second bandgap surpasses 8000 Hz. There are also two longitudinal bandgaps, designated BG1 and BG2, that lie within 10.5 kHz. The frequency ranges of these bandgaps are 2120.8–2776.4 Hz and 3730.8–10,094 Hz. In particular, the degree of attenuation in the second bandgap is much greater than in the first bandgap, as seen in Fig. 2(d). Additionally, a complete bandgap exists in the first longitudinal bandgap, which impedes the propagation of the flexural and longitudinal waves. This is not surprising, given the flexural and longitudinal bandgaps overlap in frequency bands. Furthermore, from the perspective of the vibration modes, this complete bandgap is formed by the mode transition phenomena [69], which is comparatively smaller than the previous complete bandgap, both in terms of bandwidth and imaginary wave vector. Notably, the flexural wave bandgaps BG5 and BG6 are also part of the complete bandgap, and BG4 is both a unidirectional and complete bandgap, whereas BG1 and BG2 are unidirectional bandgaps.

To study the mechanism of the flexural and longitudinal bandgaps, Fig. 3 and 4 illustrate the vibration modes of the flexural and longitudinal bandgaps at the boundary frequency, respectively. Figure 3 depicts the eigenmodes of the flexural dispersion curves at points A_1-F_2 in the band

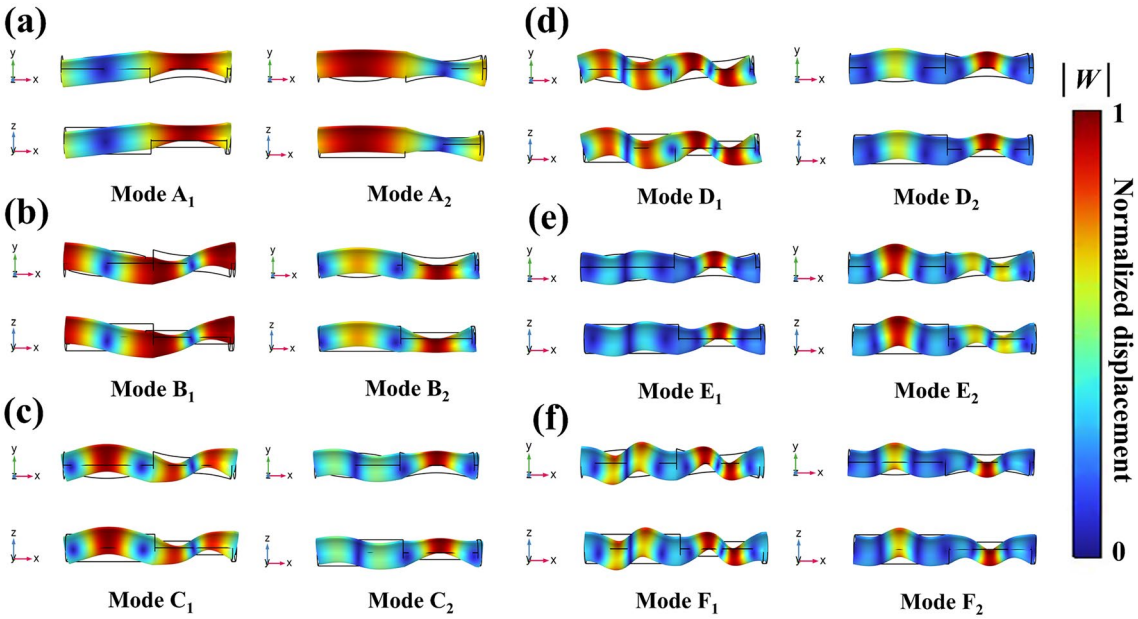
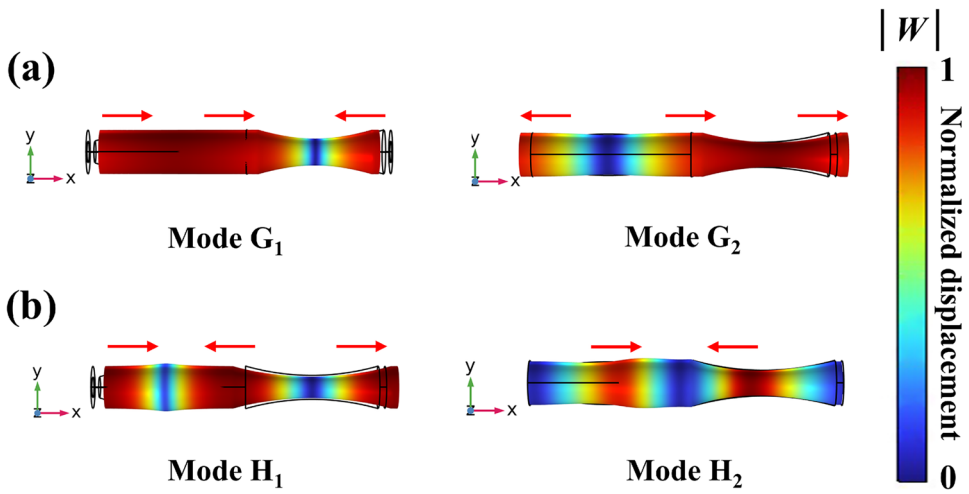


Fig. 3 **a**-**f** Normalized displacement fields of the flexural eigenmodes marked on the dispersion curves (The color distribution on the geometries indicates the magnitude of the total modal displacement.)

Fig. 4 **a** and **b** Normalized displacement fields of the longitudinal eigenmodes marked on the dispersion curves (The color distribution on the geometries indicates the magnitude of the total modal displacement.)



structure of Fig. 2(a), where the odd-numbered rows represent the y direction flexural vibration modes and the even-numbered rows represent the z direction flexural vibration modes. The vibration modes of the y direction and z direction correspond to the same eigenfrequencies due to the symmetry of the beam structure. Additionally, the y and z flexural modes are orthogonal.

Before delving into the underlying mechanisms that induce bandgaps, it is important to recall that two cut-off frequencies characterize the ABH effect. The first cut-off frequency, denoted as $f_{rabh} = \frac{\pi h_2}{8h_1}\sqrt{\frac{E}{3\rho}}$, depends on the length of the ABH. The second cut-off frequency, denoted as $f_\varepsilon = \frac{\varepsilon}{2\pi}\sqrt{\frac{E}{3\rho}}$, depends on the gradient of the ABH profile. Therefore, achieving sufficient smoothness in the ABH profile is essential to minimize the reflection of incident waves. Generally $f_{abh} < f$, indicating that frequencies below f_{abh} are insufficient to activate the ABH effect. At these frequencies, the wavelength of the incident wave exceeds the width of the ABH, preventing activation of the ABH effect. When $f_{abh} < f < 3f_\varepsilon$, some waves may be partially confined and partially reflected at the ABH boundary, leading to a partial ABH effect. However, when $f > 3f_\varepsilon$, the ABH effect is fully activated. In the following, the mechanism underlying the induced bandgap is analyzed by integrating the cut-off frequencies with the resonant mode shapes depicted in Fig. 3. In the current case, $f_{abh} = 52.8$ Hz and $f_\varepsilon = 374.9$ Hz. It is observed that the lower boundary frequency of BG1 is 224.98 Hz, which is higher than f_{abh} but lower than $3f_\varepsilon$. Therefore, the modal shape at point A_1 displays strong vibrations across both the thinner and thicker regions of the inner concave ABH region, driven by the ABH effect and boundary reflections. The modal characteristics at point A_2 resemble those at A_1, with the primary difference being that the modal shape at A_2 engages with the outer convex ABH region. Therefore, it can be concluded that the first bandgap arises from localized resonance induced by the ABH effect across different regions. The upper boundary frequency of BG2 is 1405.1 Hz, which is higher than $3f_\varepsilon$. The ABH effect is fully triggered, resulting in the vibrational energy at point B_2 primarily transferring to the thinner inner concave ABH region. Meanwhile, the Bragg scattering effect becomes apparent at the lower boundary frequency of BG2, with the modal shape at point B_1 exhibiting significant vibrations across the thicker regions of both the inner concave and outer convex ABH, as well as the uniform regions. Consequently, the second bandgap is formed through the combined effects of local resonance and Bragg scattering. Similarly, the lower boundary frequency of BG3 falls within the active frequency range of the ABH effect, indicated by the vibration energy at point C_2, which is primarily concentrated in the thinner inner concave ABH region, with minimal energy in the thicker outer convex ABH region. The Bragg scattering effect becomes evident at the lower boundary frequency of BG3, with the modal shape at point C_1 showing significant vibration in the thinner outer convex ABH region and the thicker inner concave ABH region, demonstrating triple-bending vibration induced by Bragg scattering. Likewise, the third bandgap is opened as a result of the interplay between Bragg scattering and local resonance across distinct ABH regions. Similarly, analysis of modal shapes indicates that the fourth bandgap opens due to local resonance driven by both Bragg scattering and the ABH effect.

Unlike the first four band gaps, the Bragg scattering effect is less pronounced at the upper and lower boundary frequencies of BG4, whereas the ABH effect becomes more prominent, as evidenced by the modal shapes at points E_1 and E_2. Consequently, it can be concluded that the opening of the fifth bandgap is primarily attributed to local resonance driven by the ABH effect. At the lower boundary frequency of BG6, the modal shape at point F_1 resembles that at point D_1, demonstrating strong vibrations in the thicker regions of both the inner concave and outer convex ABHs. This shape is characterized by a quintuple-bending vibration, evidently due to the Bragg scattering effect. At the upper boundary frequency of BG6, the vibration energy at point F_2 concentrates in the thinner regions of both the inner concave and outer convex ABHs, indicating local resonance driven by the ABH effect. Consequently, the sixth bandgap is generated through local resonance resulting from the combined influence of Bragg scattering and the ABH effect.

Figure 4 depicts the characteristic modes of the longitudinal dispersion curves at the G_1-H_2 points in the band structure shown in Fig. 2(c). The lower boundary frequency of BG1 is 2120.8 Hz. Although it fully activates the ABH effect, the ABH effect demonstrates a unique advantage in capturing flexural waves, while its impact on longitudinal waves remains comparatively limited. Therefore, the G_1 point exhibits a longitudinal compression mode vibration pattern. The uniform region on the right side of the inner concave ABH is compressed towards the thinner region on the left side. The entire outer convex ABH region is compressed in the x positive direction. However, there is no noticeable vibration near the thinnest region of the inner concave ABH during this process. Conversely, the G_2 point shows a longitudinal stretching mode vibration pattern. The entire area of the inner concave ABH stretches in the x positive direction. The outer convex ABH stretches from the vicinity of the thinnest region in the middle to the thicker areas on both sides. However, there is no significant vibration near the thinnest region of the outer convex ABH. Consequently, it can be deduced that the Bragg scattering effect of the elastic wave at the junction of the sudden thickness shift is the main reason for the first bandgap opening.

The lower boundary frequency of BG2 is 3730.8 Hz, satisfying the conditions for activating the ABH effect.

Interestingly, regardless of whether the ABH profile is concave or convex, no noticeable vibration appears near the thinnest central region, suggesting that the lower boundary of BG2 is not strongly associated with the ABH effect. Point H_1 exhibits a tensile-compressive mode vibration pattern. The inner concave ABH stretches from the edge of the thinnest central region to the x positive and x negative directions on both sides. The outer convex ABH extrudes from the thicker, homogeneous areas on both sides toward the thinnest central region in the opposite direction. Point H_2 reveals a localized tensile-compressive modal vibration pattern. The inner concave ABH and outer convex ABH tend to be tensile-compressive from the thinner to the thicker regions, with barely any vibration in the uniform region. This observation suggests that the majority of waves concentrate at the interfaces where abrupt thickness variations occur. Consequently, the second bandgap opens for the same reason as the first bandgap.

The mechanism of the unidirectional bandgap was discussed through the total mode displacement in the preceding section. The subsequent section focuses on investigating the impact of directional displacement on the complete bandgap formation. To further explore the mechanism underlying the complete bandgap, the vibration modes at the initial endpoint of the complete bandgap are illustrated in Fig. 5(b) and (c). Due to the asymmetry in the unit cell structure, the vibration modes exhibit a high degree of complexity. It can be observed that the mode at point C_1 is mainly governed by displacement in the y direction, with displacements in the x and z directions having minimal contribution to the total displacement. Additionally, this vibration mode is strongly affected by the Bragg effect, as it exhibits characteristic global vibrations. Furthermore, the mode at point C_2 is also mainly governed by displacement in the y direction, but the displacements in the other two directions, particularly the z direction, cannot be neglected. It is observed that this vibration mode is highly influenced by the ABH effect, as it demonstrates a concentration of vibration energy in the thinner concave ABH region. From the explanation of the mode at point C_2 above, it is evident that the ABH effect has been fully triggered, leading to wave trapping. Therefore, this complete bandgap arises from the coupling effect of the ABH effect and the Bragg effect. Undeniably, this bandgap is both a complete bandgap and a flexural bandgap.

Meanwhile, a unidirectional bandgap is observed near the complete bandgap. To explain its formation mechanism, this paper examines two adjacent longitudinal bandgaps in proximity to the first complete bandgap as examples. These bandgaps are in the ranges of 2120.8–2401.3 Hz and 2726.4–2776.4 Hz. The corresponding modal characteristics at the boundaries are presented in Fig. 5(a) and (d). It can be observed that the mode at point G_1 is mainly governed by displacement in the x direction, with displacements in the other two directions making minimal contributions to the total displacement, and these displacements are localized in the thinner inner concave ABH region. Therefore, the mode characteristic at point G_1 is identified as a longitudinal compressive vibration pattern induced by the Bragg effect. In fact, although the physical mechanisms of the modes at points C_1 and G_1 share similarities, their dominant directions

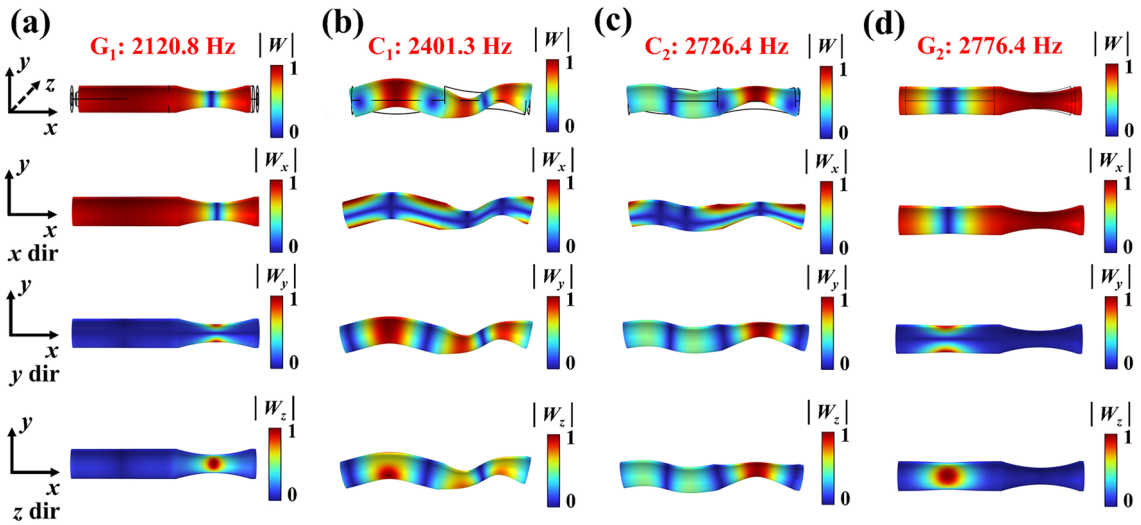


Fig. 5 **a** Eigenmodes G_1 at lower limit 2120.8Hz of longitudinal bandgap with normalized displacement value; **b** eigenmodes C_1 at lower limit 2401.3Hz of complete bandgap with normalized displacement value; **c** eigenmodes C_2 at upper limit 2726.4Hz of complete bandgap with normalized displacement value; **d** eigenmodes G_2 at upper limit 2776.4Hz of longitudinal bandgap with normalized displacement value (The first row is total modal displacement value, second row is x direction component, third row is y direction component and fourth row is z direction component. The color distribution on the geometries indicates the magnitude of the total modal displacement.)

are distinct. The transition from y direction dominance to x direction dominance is the primary factor contributing to the formation of the unidirectional longitudinal bandgap. Due to the bidirectional nature of the structure, the mode characteristic at G_2 contrasts with that of G_1. The most notable feature is that the displacements in the y and z directions are primarily localized in the thinner outer convex ABH region. Similarly, the formation of the second directional bandgap in the x direction correlates with the transformation of the dominant displacement. It can be concluded that the formation of the complete bandgap stems from the transformation of the longitudinal bandgap into a flexural bandgap. Additionally, since there is a direct transformation of vibration modes preceding and following the complete bandgap, mode transformation phenomena are also evident.

Vibration Transmission Spectra of Bidirectional Asymmetric ABH Metamaterial Beam

The above analysis shows that the bidirectional asymmetric ABH metamaterial beam filtering characteristics are reflected by the bandgap of an infinite period. However, the size of the actual structure is always finite, and the attenuation factor of the evanescent wave at the bandgap frequency is affected in practical applications.

In this section, the finite element method (FEM) is used to calculate the vibration transmission rate in a finite structure. To quantitatively demonstrate the wider attenuation region and enhanced attenuation effect of the bidirectional asymmetric ABH beams proposed in this paper, the single concave and single convex ABH beams are used as a control group, and the number of periods of the finite structure is set to four, as shown in Fig. 6. Furthermore, some geometrical parameters and material properties of the various structures were kept consistent in order to make a valid comparison between them. The flexural and longitudinal excitation boundary loads were applied on the small end face, followed by the excitation and response probe points on the small and large end faces. The structure operates under free boundary conditions. The vibration transmission loss is defined as follows:

$$\text{Transmission}\,(dB) = 20\log_{10}\left(\left|\frac{w_{out}}{w_{in}}\right|\right) \tag{9}$$

where W_{out} and W_{in} represent the displacement amplitudes at the response and excitation locations, respectively. The transmission loss results are depicted in Fig. 7.

According to Fig. 7(b) and (d), the vibration transmission loss of the bidirectional asymmetric ABH metamaterial beam is over 40 dB in the flexural direction and 100 dB in the longitudinal direction, consistent with the calculated flexural bandgap and longitudinal bandgap. In contrast, with single convex ABH metamaterial beams, there is no significant vibration damping effect observed in the flexural direction, and transmissibility decreases at certain frequencies. The vibration reduction frequency band for longitudinal waves is narrow, exhibiting a much lower attenuation effect

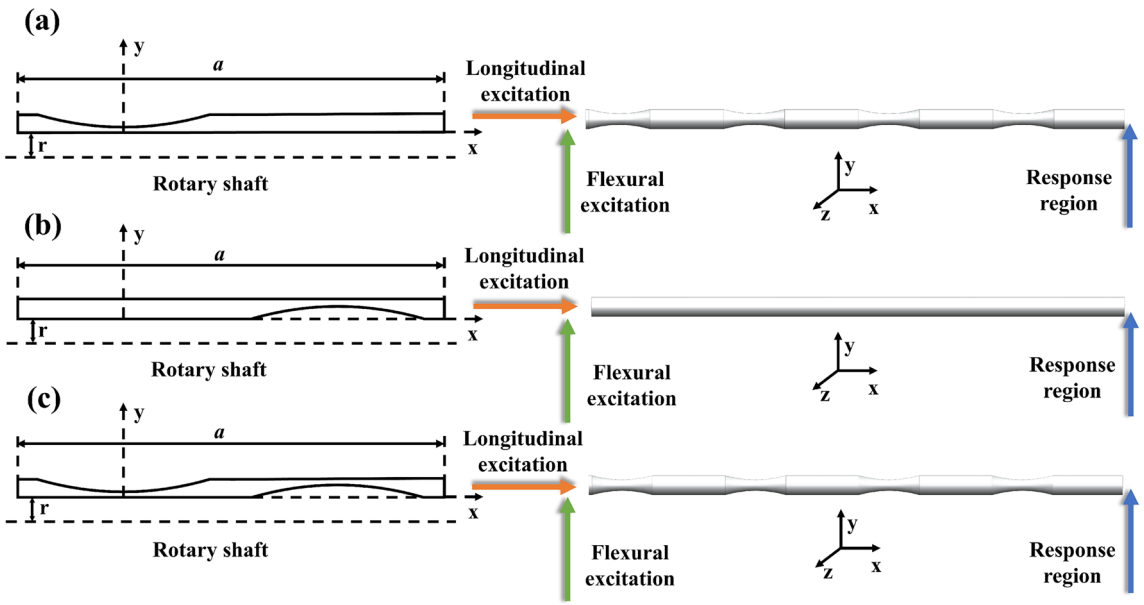


Fig. 6 **a** Finite period single concave ABH beam model; **b** finite period single convex ABH beam model; **c** finite period bidirectional asymmetric ABH beam model. The number of cycles is four. The endpoints of the finite structure are referred to as the excitation and response zones (The left side depicts the thickness profile of a single cell from the matching model on the right side; orange, green, and blue arrows represent the longitudinal excitation, flexural excitation, and response zones, respectively)

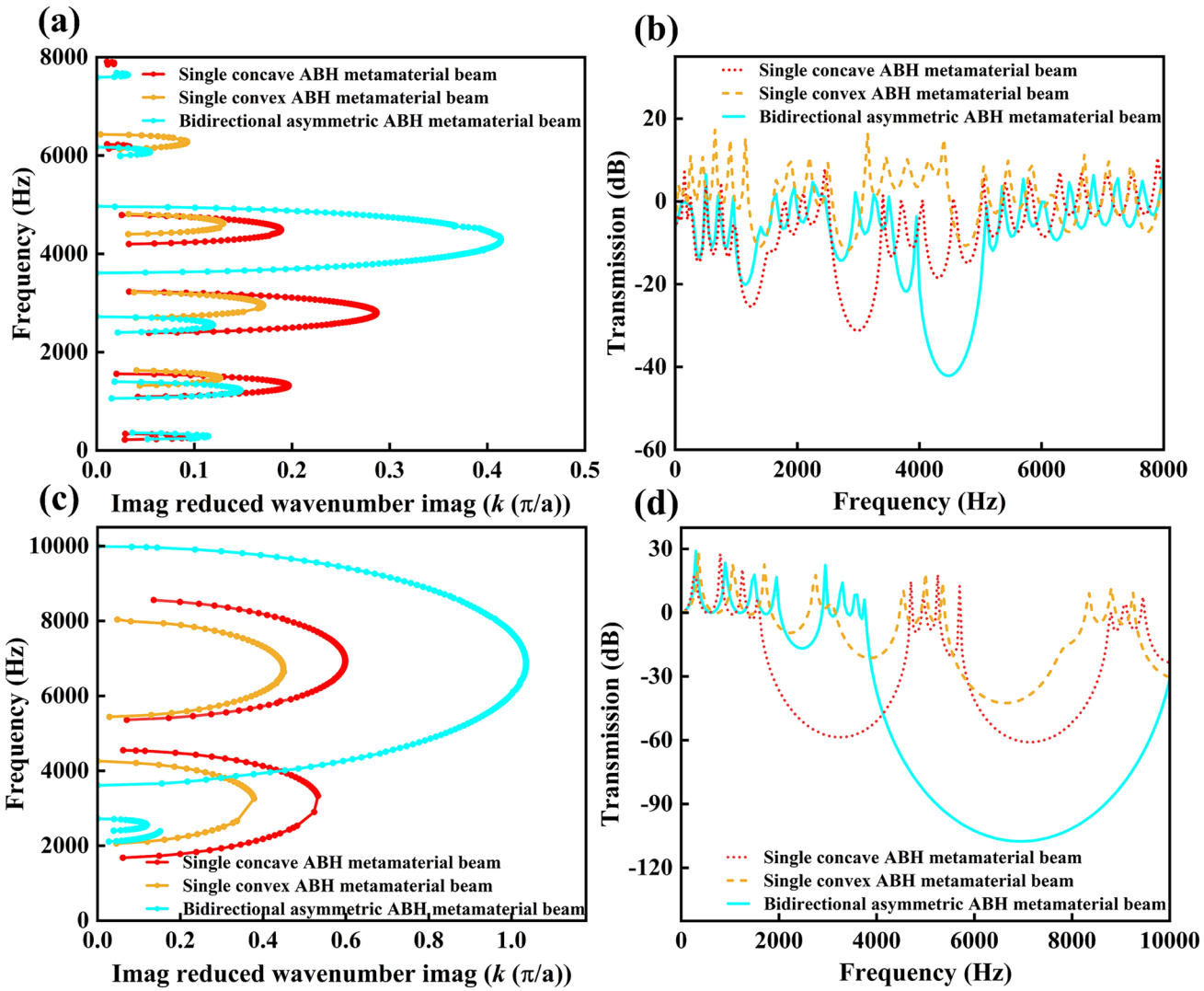


Fig. 7 **a** and **c** Flexural and longitudinal imaginary part dispersion curves of various configurations of ABH metamaterial beam calculated by the PDE $k(\omega)$ method, respectively; **b** and **d** flexural and longitudinal vibration transfer curves derived by FEM for various configurations of ABH metamaterial beam, respectively

compared to the bidirectional asymmetric type. Conversely, the single concave type exhibits a significantly larger vibration-damping effect in the flexural direction than the single convex type. It shows a slightly higher transmission loss than the bidirectional asymmetric type in certain frequency bands. In the longitudinal direction, the single concave type demonstrates a wider frequency range and higher transmission loss than the single convex type. However, the bidirectional asymmetric type demonstrates a more significant vibration reduction effect compared to the single concave type, with an increase in transmission loss of approximately 67%.

According to the complex band results, imaginary wave vectors can directly characterize wave attenuation performance. Larger values of imaginary wave vectors indicate stronger attenuation. As a result, the imaginary parts of the control group complex band are calculated, displayed, and analyzed, compared with those of the new structure provided in this study. Figure 7(a) and (c) illustrate the attenuation factors of the evanescent wave in the flexural and longitudinal directions for three distinct structures: single concave, single convex, and bidirectional asymmetric. The results show that the flexural attenuation factor of the bidirectional asymmetric type is discretely distributed within the studied frequency band, and that the attenuation factor is greater than those of the single concave and single convex structures in certain frequency ranges. The single convex structure exhibits the smallest degree of attenuation, corresponding to the vibrational transmission loss diagram in Fig. 7(b). Meanwhile, the longitudinal attenuation factor of the bidirectional

asymmetric type is focused within a specific frequency range, achieving approximately 66.8% vibration reduction in the studied frequency range. Moreover, the magnitude of the attenuation factor indicates the highest level of attenuation, consistent with the results shown in Fig. 7(d). All these findings suggest that the bandgap characteristics are stable within the finite structure of the bidirectional asymmetric ABH metamaterial beam.

The above analysis is based on results obtained without accounting for the material damping factor. In real-world structures, material damping factors are inherent, and different materials demonstrate varying damping properties. To investigate the effect of the material damping factor on bandgap attenuation performance, vibration transmission loss results for various damping factors were plotted, as illustrated in Fig. 8. To better illustrate the effect of the damping factor on attenuation, for both flexural and longitudinal waves, we use the vibration transmission loss of -20 dB under no damping as a reference. It should be noted that the range of damping factors η considered in this study spans from 0.01 to 0.1, with increments of 0.01. As the damping factor increases, frequency ranges of sharp attenuation for flexural waves emerge, and the number of these attenuation ranges also increases, as illustrated in Fig. 8(a). Furthermore, considering the effect of different damping factors on flexural bandwidth, it can be concluded that while the damping factor has a minor impact on bandwidth, it enhances attenuation capability within the frequency ranges, which aligns with expectations. Similarly, as shown in Fig. 8(b), with increasing damping factor, vibration transmission loss in most areas of the longitudinal wave shows no significant change, with only a small portion exhibiting weakened attenuation. Overall, the damping factor has a minimal effect on longitudinal wave attenuation.

Method of Tuning the Bandgap

Before examining the effect of geometric parameters on bandgap performance, we first analyze the influence of the damping factor of the material on its bandgap characteristics. Figure 9 illustrates the effects of varying damping factors η on the bandgap performance. As depicted in Fig. 9(a) and (c), the upper and lower boundary frequencies of the flexural and longitudinal bandgaps exhibit no significant changes as the damping factor increases. Likewise, the flexural and longitudinal bandwidths remain almost unchanged, as shown in Fig. 9(b) and (d). Thus, it can be concluded that the damping factor has a minimal effect on bandgap performance. To investigate how the proposed structure regulates bandgap performance, we assume the material has no damping.

Figure 10 depicts the effect of geometric parameters on the flexural wave bandgap of a bidirectional asymmetric ABH metamaterial beam. As illustrated in Fig. 10(a), the power index $m = 2,3,4,5,6,7,8,9$ and 10 were chosen to explore the effect of the power index on the six bandgaps. It is evident that, except for the bandwidth slightly larger than BG4, increasing the power index leads to a lower boundary frequency. Simultaneously, the upper boundary frequencies of the first, fifth, and sixth bandgaps shift to lower frequencies. The fifth and sixth bandwidths increase and decrease, while the first bandwidth remains unchanged. The second and third bandwidths show a different tendency, which is closely linked to the trend of the upper boundary frequency. Similar to the conventional symmetric ABH, a larger power-law index results in lower structural stiffness, leading to a lower frequency range for the bandgap. However, the absolute bandwidth does not show a single upward or downward trend but rather exhibits an ordered mix of rises and falls. Furthermore, increasing the power index will steepen the

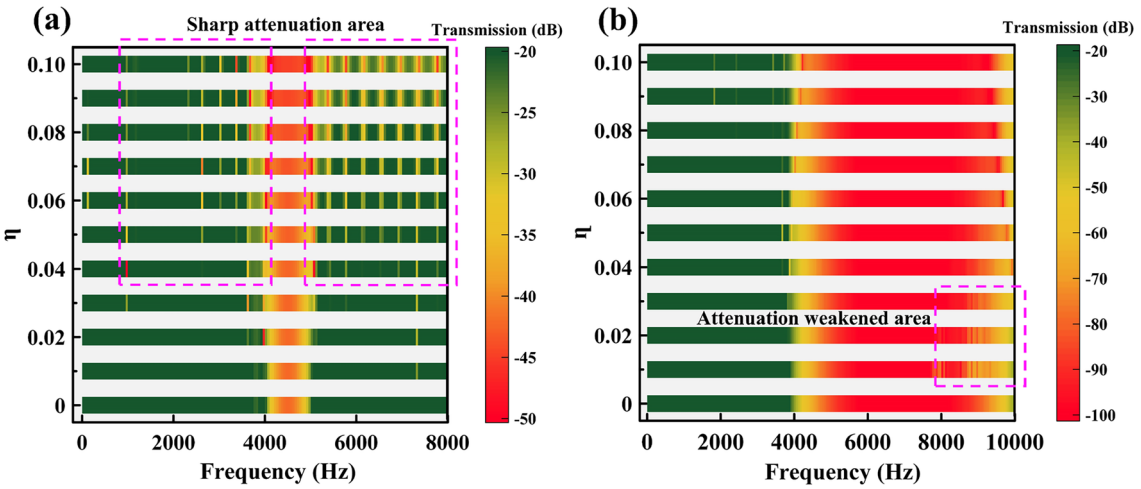


Fig. 8 Effect of different damping factors η on bandgap attenuation performance; **a** flexural waves; **b** longitudinal waves

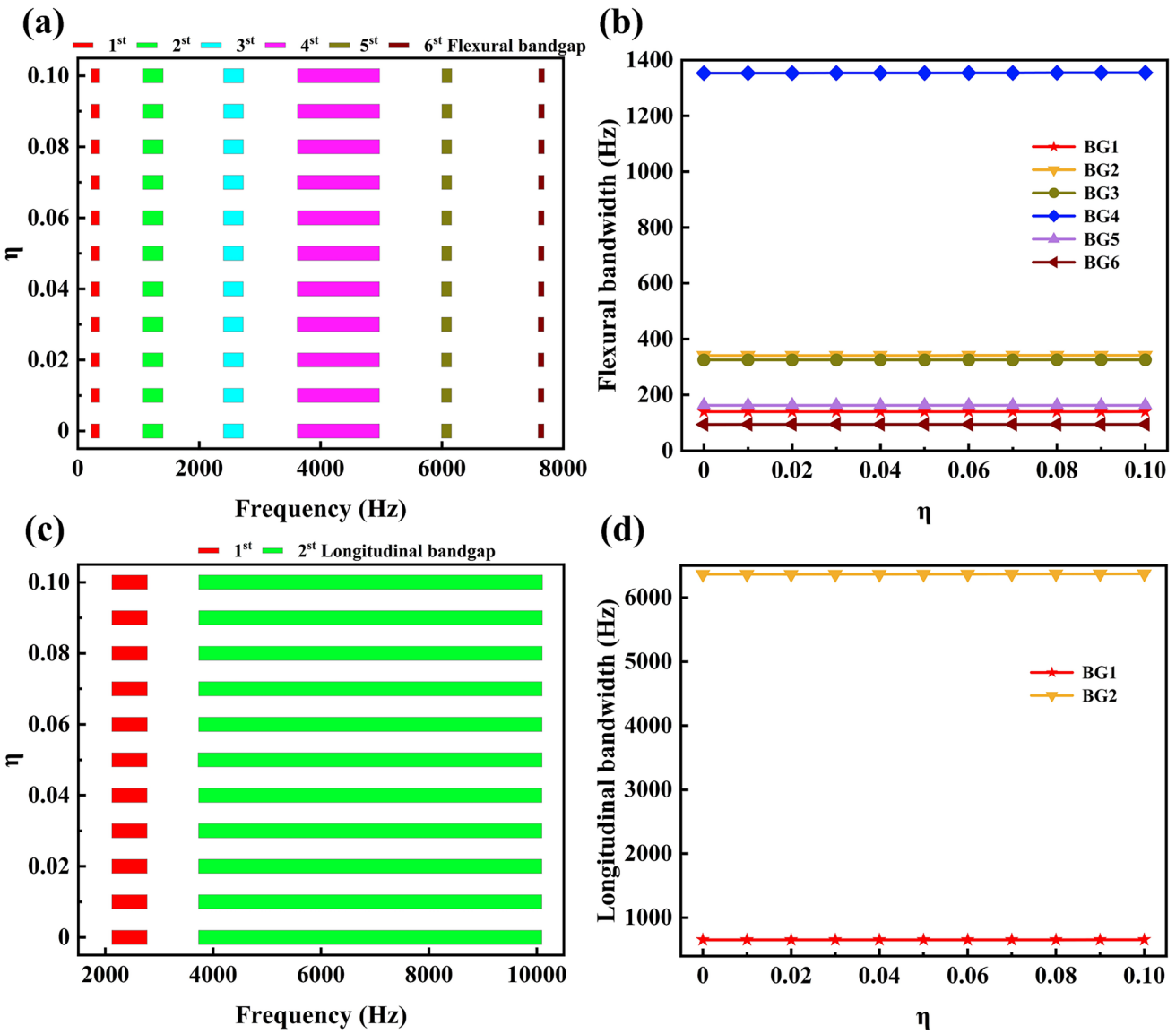


Fig. 9 Effect of different damping factors η on bandgap performance; **a** flexural bandgap; **b** flexural bandwidth; **c** longitudinal bandgap; **d** longitudinal bandwidth

slope of the ABH profile, resulting in a weaker ABH effect and thus more reflection of the incident wave at the ABH, leading to a smaller partial bandwidth. However, the asymmetric ABH structure at the junction of the thickness mutation will benefit the Bragg scattering effect, increasing the partial bandwidth. As displayed in Fig. 10(b), rotary shafts with $r = 2,3,4,5,6,7,8,9$ and 10 mm are used to study the influence of rotary shafts on the six bandgaps. It can be concluded that increasing the rotary shaft radius causes a higher frequency range, with a significant decrease in the third and fifth bandwidths and a significant increase in the fourth bandwidth. This is attributed to the increase in the overall size of the structure initiated by the raised rotary shaft, where the Bragg scattering effect plays a dominant role.

As observed in Fig. 10(c), the residual thicknesses h_1 were selected to investigate the effect of residual thickness on the upper and lower boundary frequencies of the six bandgaps. It can be seen that the upper and lower boundary frequencies converge as the residual thickness increases. This is due to the fact that with the increase of the residual thickness, the ABH effect decreases, resulting in the weakening of the localized resonance. It should also be noted that when h_1 is sufficiently large, all the waves of the bandgap will disappear, as shown in Fig. 10(d). Additionally, Fig. 10(e) reveals the effect of ABH length on the upper and lower boundary frequencies of the six bandgaps. The increase of ABH length has less effect on the lower boundary frequencies of the first, second, and third bandgaps, and more effect

on the lower boundary frequencies of the fourth and sixth bandgaps. The latter shows a tendency to increase the lower boundary frequencies. The upper boundary frequencies of the second and third bandgaps increase with the increase in ABH length, while the upper boundary frequency of the sixth bandgap exhibits the opposite behavior. The upper boundary frequencies of the fourth and fifth bandgaps tend to climb and subsequently fall. Additionally, Fig. 10(f) illustrates that the ABH length has a minor effect on the width of the first bandgap but a more significant effect on the widths of the fourth and sixth bandgaps, with the greatest bandwidth of the fourth bandgap occurring at $l_1 = 30$mm. As the length of ABH increases, so do the widths of the second and third bandgaps. The bandwidth of the fifth bandgap tends to expand before decreasing and then increasing again, with the bandwidth virtually disappearing at $l_1 = 45$mm.

In summary, geometric parameters significantly impact the flexural wave bandgap, and a well-designed set of geometric parameters can result in the adjustable bandgap required for bidirectional asymmetric ABH metamaterial beams in practical vibration isolation scenarios.

Figure 11 depicts the impact of geometric parameters on the longitudinal bandgap of bidirectional asymmetric ABH metamaterial beams. Figure 11(a) shows that as the power index m increases, the upper boundary frequency of the second bandgap decreases, and the lower boundary frequency changes less, which leads to a smaller absolute bandwidth. Figure 11(b) reflects the effect of the difference in the rotary shaft r on the bandgap, demonstrating that as the rotary shaft r increases, the upper boundary frequencies of the first and second bandgaps shift to lower frequencies. The lower boundary frequency of the first bandgap shifts to a higher frequency, and the lower boundary frequency of the second bandgap does not change considerably, resulting in a decrease in the absolute bandwidth. Figure 11(c) reveals that as the residual thickness h_1 increases, the upper and lower boundary frequencies of the first bandgap shift to higher frequencies, and the rate of increase of the upper boundary frequency is slower than that of the lower boundary frequency, resulting in the upper and lower boundary frequencies converging. The upper boundary frequency of the second bandgap shifts to a lower frequency, whereas the lower boundary frequency shifts to a higher frequency. It can be observed from Fig. 11(d) that the bandwidths of the first and second bandgaps decrease as the residual thickness h_1 increases, and it should be noted that the longitudinal bandgap will disappear when h_1 is in adequate proximity to the homogeneous thickness h_2 because the Bragg scattering effect is not at work. As displayed in Fig. 11(e), as the ABH length increases, the effect on the upper and lower boundary frequencies of the first bandgap is small, which is due to the slight impact of the ABH effect on the longitudinal bandgap. However, the upper boundary frequency of the second bandgap shows a tendency to rise and then fall as the ABH length increases, while the lower boundary frequency shows a tendency to fall and then rise. Combined with Fig. 11(f), it can be observed that the bandwidth of the first bandgap increases slightly as the ABH length increases, but the overall change is not significant. The bandwidth of the second bandgap tends to increase and then decrease, with the maximum bandwidth occurring at $l_1 = 45$mm.

It is clear that the geometric parameter h_1 has the greatest influence on the longitudinal bandgap, and a suitable residual thickness h_1 design can result in a stable longitudinal bandgap for bidirectional asymmetric ABH metamaterial beams.

Experimental Study of Bidirectional Asymmetric ABH Metamaterial Beam

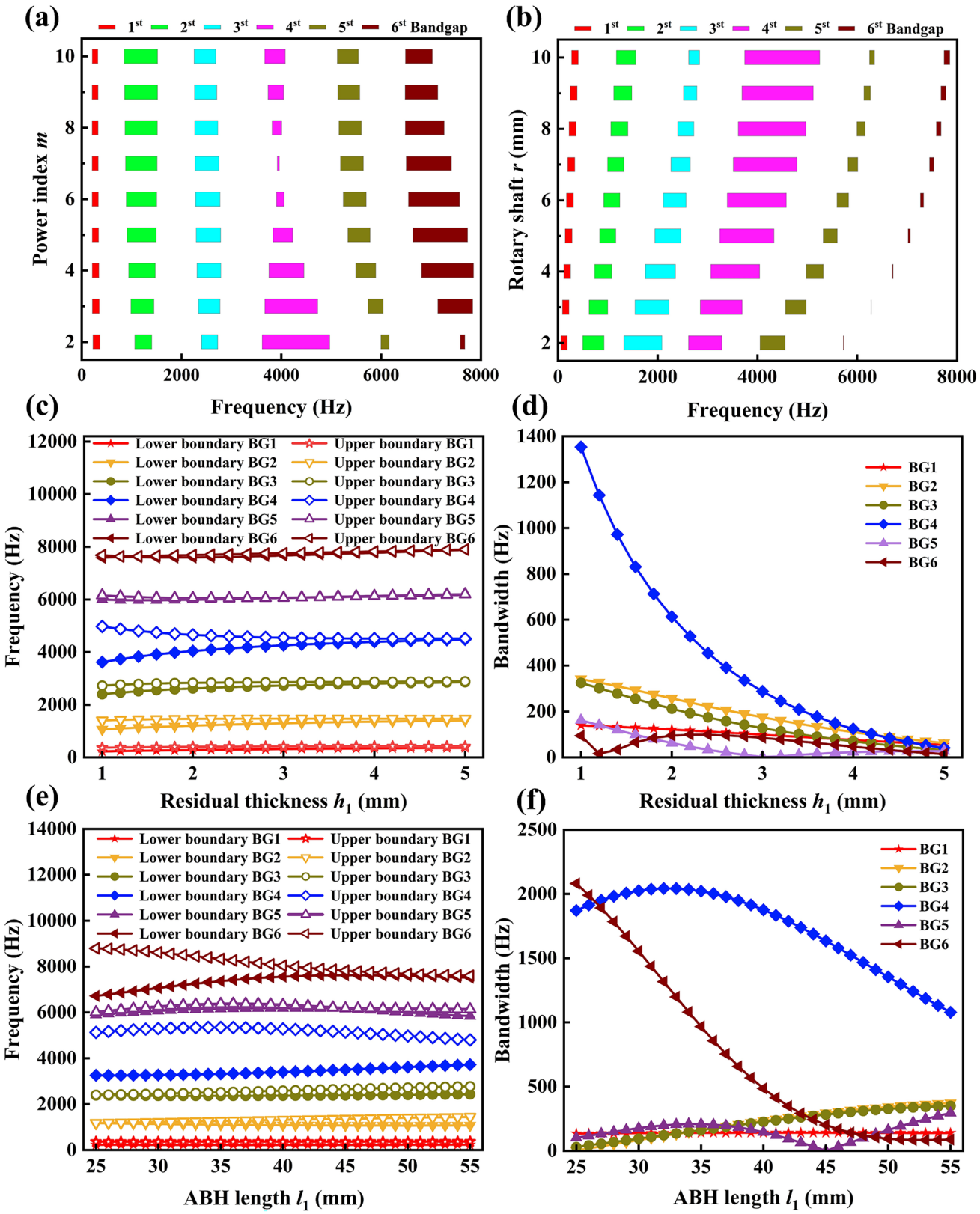
To verify the bandgap and vibration damping performance of bidirectional asymmetric ABH metamaterial beams, this section includes experiments and discusses the transmissibility of flexural and longitudinal vibrations.

Experimental Setup

The bidirectional asymmetric ABH metamaterial beam is created using a 3D printing approach, maintaining the same shape and material characteristics described in Sect. "Band structure and mechanisms" for the numerical analysis. Figure 12(a) and (b) show the vibration test flowchart and the primary test equipment, respectively. The experimental samples are supported on the support frame by two flexible nylon ropes, which are utilized to model the free boundary conditions identical to those employed in the simulation. During dynamic testing, the excitation mode of the signal generator was pre-set to sinusoidal sweeping, with a sweeping range of 1–12.8 kHz and a sweeping speed of 12.8 kHz/s. The sinusoidal vibration signal was then amplified by a power amplifier (B&K Power Amplifier Type 2732) and delivered to the shaker, while an electromagnetic oscillator (B&K Modal Exciter Type 4824) excited the end face boundary of the sample via an auxiliary flexible thin rod. The shaker can be used to apply flexural and longitudinal excitation to the small end faces of the test specimen. Unidirectional accelerometers (B&K model 4507Bx) are mounted at both ends of the specimen to collect the input and output vibration signals. These signals are fed back to the computer software B&K connect system for post-processing through FFT. The transmission spectrum in decibels (dB) is calculated as follows:

$$\text{Transmission} = 20\log_{10}\left(A_{out}/A_{in}\right) \tag{10}$$

where A_{out} and A_{in}, the output and input accelerations, respectively, can be obtained using the data processing software LABSHOP.


(a)
1st 2st 3st 4st 5st 6st Bandgap
Power index m
Frequency (Hz)
(b)
1st 2st 3st 4st 5st 6st Bandgap
Rotary shaft r (mm)
Frequency (Hz)
(c)
Lower boundary BG1
Upper boundary BG1
Lower boundary BG2
Upper boundary BG2
Lower boundary BG3
Upper boundary BG3
Lower boundary BG4
Upper boundary BG4
Lower boundary BG5
Upper boundary BG5
Lower boundary BG6
Upper boundary BG6
Frequency (Hz)
Residual thickness h1 (mm)
(d)
BG1
BG2
BG3
BG4
BG5
BG6
Bandwidth (Hz)
Residual thickness h1 (mm)
(e)
Lower boundary BG1
Upper boundary BG1
Lower boundary BG2
Upper boundary BG2
Lower boundary BG3
Upper boundary BG3
Lower boundary BG4
Upper boundary BG4
Lower boundary BG5
Upper boundary BG5
Lower boundary BG6
Upper boundary BG6
Frequency (Hz)
ABH length l1 (mm)
(f)
BG1
BG2
BG3
BG4
BG5
BG6
Bandwidth (Hz)
ABH length l1 (mm)

◂**Fig. 10** Effect of various geometrical parameters on the flexural bandgap of a metamaterial beam with bidirectional asymmetric ABH; **a** effect of the power index $m \in [2, 3, 4, 5, 6, 7, 8, 9, 10]$ on the bandgap; **b** effect of the rotary shaft position $r \in [2, 3, 4, 5, 6, 7, 8, 9, 10]$ on the bandgap; **c** effect of the residual thickness h_1 on the frequencies of the upper and lower bandgap borders; **d** effect of residual thickness h_1 on the bandgap bandwidth; **e** effect of ABH length l_1 on the bandgap upper and lower boundary frequencies; **f** effect of ABH length l_1 on the bandgap bandwidth

Experimental Results and Discussion

As seen in Fig. 13, vibration transfer studies were conducted in the flexural and longitudinal directions, respectively. The discrepancies at higher frequencies may be primarily caused by dimensional machining accuracy, real boundary conditions, and the torsional waves omitted in the numerical simulation, which are more visible at higher frequencies. It was discovered that there are apparent attenuation bands in the transmission spectrum in both directions, and the flexural wave attenuation corresponds well to the first four band gaps. However, it fails to match well with the last two bandgaps due to the small flexural attenuation factor in these two bandgaps, the large loss factor of the material, and the lack of dimensional precision. The longitudinal wave attenuation was the same as the bandgap, exhibiting significant broadband vibration damping. Understandably, the shallow values of vibration transmission loss predicted numerically (close to 110 dB) are difficult to achieve experimentally due to the fragile vibration levels in the attenuation bands and the unavoidable presence of torsional waves. These waves are present experimentally but were ignored in the numerical simulations. Additionally, the sample has a longer dimension in the longitudinal direction, and structural deformation is an important factor affecting the results. Overall, the experimental results generally agree with the numerical results.

Conclusion

This research proposed a unique structure, the bidirectional asymmetric ABH metamaterial beam. The flexural and longitudinal bandgaps and vibration attenuation capabilities were examined using complex band structure and dynamics analysis. To study the mechanism of bandgap formation, the vibration modes of the flexural and longitudinal bandgaps at the boundary frequencies were displayed independently. Additionally, to analyze the vibration transmission rate of the new structure over a finite period, the vibration transmission curves in the flexural and longitudinal directions were plotted individually and compared to those of typical unidirectional structural beams. Based on these findings, the following conclusions were drawn:

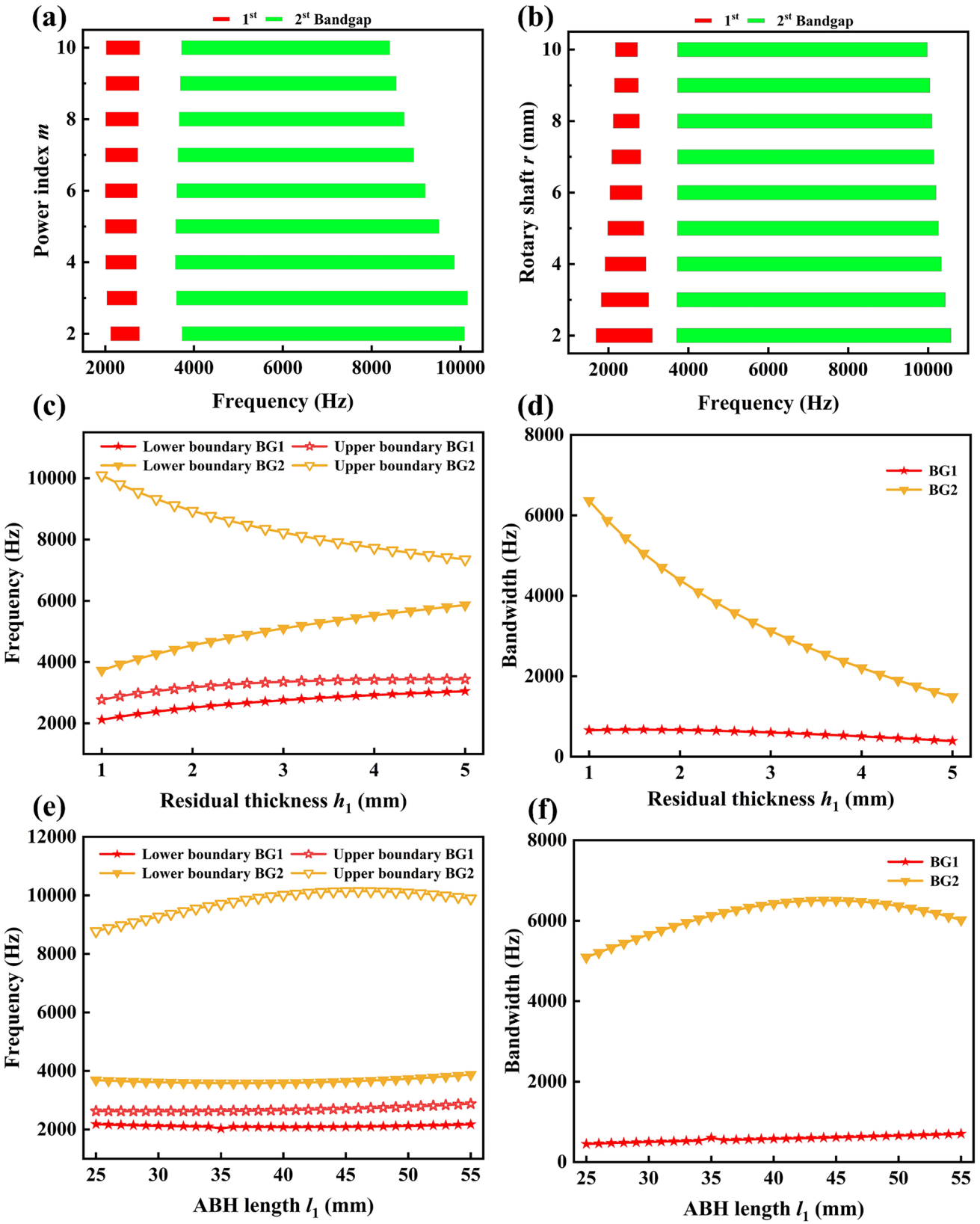
According to the complex band analysis, bidirectional asymmetric ABH metamaterial beams have greater flexural and longitudinal bandgaps than unidirectional structured beams and exhibit higher attenuation factors for evanescent waves.

For the physical mechanism of flexural bandgap formation, the coupling effect between local resonance and Bragg scattering resulting from ABH energy aggregation plays a pivotal role in bandgap opening. However, the influence of the ABH effect is less pronounced for the longitudinal bandgap. Bragg scattering is dominant because elastic waves tend to induce reflection and interference at junctions of abrupt thickness changes. The vibrational mode transformation phenomenon reveals that, in addition to unidirectional bandgaps, two complete bandgaps with large bandwidths exist in the investigated frequency band. The generation of complete bandgaps is closely related to the coupling and separation of modes between Bragg scattering and local resonance.

Generally, the bands of the bidirectional asymmetric ABH metamaterial beam have a sizeable unidirectional bandgap and several relatively large complete bandgaps. Finite element simulations also reveal that the attenuation area in the vibration transfer curve of the finite period structure corresponds to the predicted bandgap properties. Moreover, an increase in the material damping factor significantly improves the capacity for flexural wave attenuation. Furthermore, the effect on unidirectional bandgap characteristics when the geometrical parameters are varied in the same order is discussed in detail. By increasing the power-law exponent and decreasing the length of the ABH, the flexural bandgap exhibits diversity, while the longitudinal bandgap shows a smaller absolute bandwidth. Decreasing the position of the rotary shaft lowers the frequency range for flexural bandgaps and increases the absolute bandwidth for longitudinal bandgaps. Increasing the residual thickness reduces the local resonance effect for the flexural bandgap and the Bragg scattering effect for the longitudinal bandgap, adversely affecting the expanded bandgap.

To validate the numerical analysis, experimental photosensitive resin samples of bidirectional asymmetric ABH metamaterial beams were fabricated and 3D printed. There is good agreement between experimental data and numerical predictions of the bandgap. Moreover, the vibration transfer capabilities in both flexural and longitudinal directions demonstrate superior vibration attenuation performance.

The bidirectional asymmetric hierarchical structure proposed in this study breaking the limitations of spatial symmetry, providing a novel structural design reference for achieving broadband vibration reduction. Additionally, a comprehensive analysis in this paper reveals that the ABH metamaterial beam formed by this hierarchical structure exhibits not only broader unidirectional bandgap performance and stronger vibration attenuation capacity but also


(a)
1st 2st Bandgap
Power index m
Frequency (Hz)
(b)
1st 2st Bandgap
Rotary shaft r (mm)
Frequency (Hz)
(c)
Lower boundary BG1
Upper boundary BG1
Lower boundary BG2
Upper boundary BG2
Frequency (Hz)
Residual thickness h1 (mm)
(d)
BG1
BG2
Bandwidth (Hz)
Residual thickness h1 (mm)
(e)
Lower boundary BG1
Upper boundary BG1
Lower boundary BG2
Upper boundary BG2
Frequency (Hz)
ABH length l1 (mm)
(f)
BG1
BG2
Bandwidth (Hz)
ABH length l1 (mm)

◂**Fig. 11** Effect of various geometrical parameters on the longitudinal bandgap of a metamaterial beam with bidirectional asymmetric ABH; **a** effect of the power index $m \in [2, 3, 4, 5, 6, 7, 8, 9, 10]$ on the bandgap; **b** effect of the rotary shaft position $r \in [2, 3, 4, 5, 6, 7, 8, 9, 10]$ on the bandgap; **c** effect of the residual thickness h_1 on the frequencies of the upper and lower bandgap borders; **d** effect of residual thickness h_1 on the bandgap bandwidth; **e** effect of ABH length l_1 on the bandgap upper and lower boundary frequencies; **f** effect of ABH length l_1 on the bandgap bandwidth

possesses the ability to open a complete bandgap. In future work, different types of hierarchical structures could be considered to thoroughly explore the effects on vibration suppression performance, with potential for practical engineering applications.

Appendix A

PDE module in Comsol in complex band analyses

Numerical analyses of the complex band are implemented by utilizing the Partial Difference Equation (PDE) module in the commercial finite element software Comsol Multiphysics. The standard eigenvalue control equation in the module can be defined by the 3D PDE coefficient as:

$$\Lambda^2 e_a \cdot \mathrm{U} - \Lambda \mathrm{d_a} \cdot \mathrm{U} + \Delta \cdot (-\mathrm{C} : \nabla \mathrm{U} - \alpha \cdot \mathrm{U} + \gamma) + \beta : \nabla \mathrm{U} + \mathrm{a} \cdot \mathrm{U} = \mathrm{f} \tag{11}$$

where Λ denotes the scalar eigenvalue; $e_a, d_a, C, \alpha, \beta$ and a are the mass coefficient, damping coefficient, diffusion coefficient, conservative flux convection coefficient, convection coefficient and absorption coefficient, respectively;U, γ and f are represented by the eigenvector, conservative flux source and source terms, respectively. These parameters are shown in detail as:

$$e_a = \begin{bmatrix} -(\lambda+2\mu) & 0 & 0 \\ 0 & -\mu & 0 \\ 0 & 0 & -\mu \end{bmatrix}, d_a = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$C = \begin{bmatrix} \begin{bmatrix} \lambda+2\mu & 0 & 0 \\ 0 & \mu & 0 \\ 0 & 0 & \mu \end{bmatrix} & \begin{bmatrix} 0 & \lambda & 0 \\ \mu & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} & \begin{bmatrix} 0 & 0 & \lambda \\ 0 & 0 & 0 \\ \mu & 0 & 0 \end{bmatrix} \\ \begin{bmatrix} 0 & \mu & 0 \\ \lambda & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} & \begin{bmatrix} \mu & 0 & 0 \\ 0 & \lambda+2\mu & 0 \\ 0 & 0 & \mu \end{bmatrix} & \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \lambda \\ 0 & \mu & 0 \end{bmatrix} \\ \begin{bmatrix} 0 & 0 & \mu \\ 0 & 0 & 0 \\ \lambda & 0 & 0 \end{bmatrix} & \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \mu \\ \lambda & 0 & 0 \end{bmatrix} & \begin{bmatrix} \mu & 0 & 0 \\ 0 & \mu & 0 \\ 0 & 0 & \lambda+2\mu \end{bmatrix} \end{bmatrix} \tag{12}$$

$$\alpha = \begin{bmatrix} \begin{bmatrix} -(\lambda+2\mu)\Lambda \\ 0 \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ -\mu\Lambda \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \\ -\mu\Lambda \end{bmatrix} \\ \begin{bmatrix} 0 \\ -\lambda\Lambda \\ 0 \end{bmatrix} & \begin{bmatrix} -\mu\Lambda \\ 0 \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 0 \\ 0 \\ -\lambda\Lambda \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} & \begin{bmatrix} -\mu\Lambda \\ 0 \\ 0 \end{bmatrix} \end{bmatrix}$$

and

$$\beta = \begin{bmatrix} \begin{bmatrix} (\lambda+2\mu)\Lambda \\ 0 \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ \lambda\Lambda \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \\ \lambda\Lambda \end{bmatrix} \\ \begin{bmatrix} 0 \\ \mu\Lambda \\ 0 \end{bmatrix} & \begin{bmatrix} \mu\Lambda \\ 0 \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 0 \\ 0 \\ \mu\Lambda \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} & \begin{bmatrix} \mu\Lambda \\ 0 \\ 0 \end{bmatrix} \end{bmatrix}, \gamma = \begin{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \end{bmatrix}$$

$$a = \begin{bmatrix} -\rho\omega^2 & 0 & 0 \\ 0 & -\rho\omega^2 & 0 \\ 0 & 0 & -\rho\omega^2 \end{bmatrix}, f = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \tag{13}$$

where λ and μ are the first and second Lame constants expressed as $\lambda = \frac{\nu E}{(1-2\nu)(1+\nu)}$ and $\mu = \frac{E}{2(1+\nu)}$, respectively. By substituting Eqs. (12, 13) into Eq. (11), the eigenvalues Λ can be calculated and the complex wavenumber is obtained by $k = -i\Lambda$.

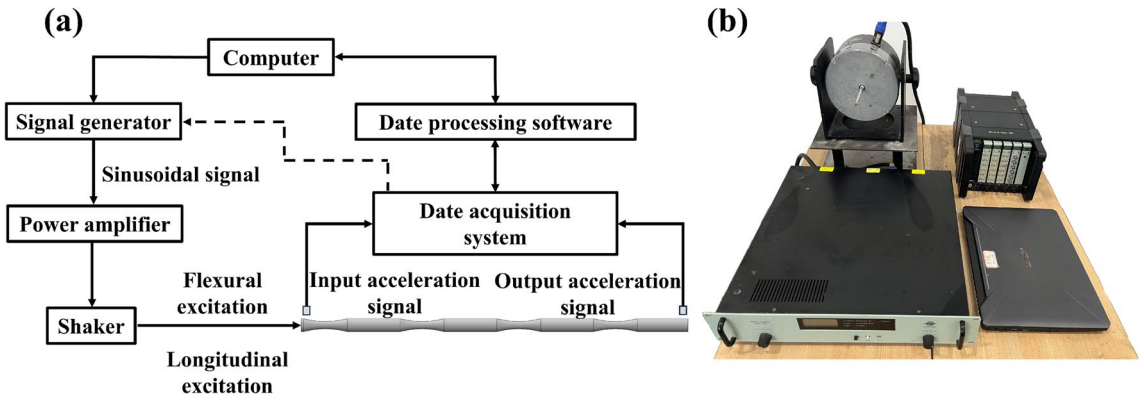


Fig. 12 **a** Flowchart of the experimental test with excitation and response points; **b** photographs of the primary test equipment

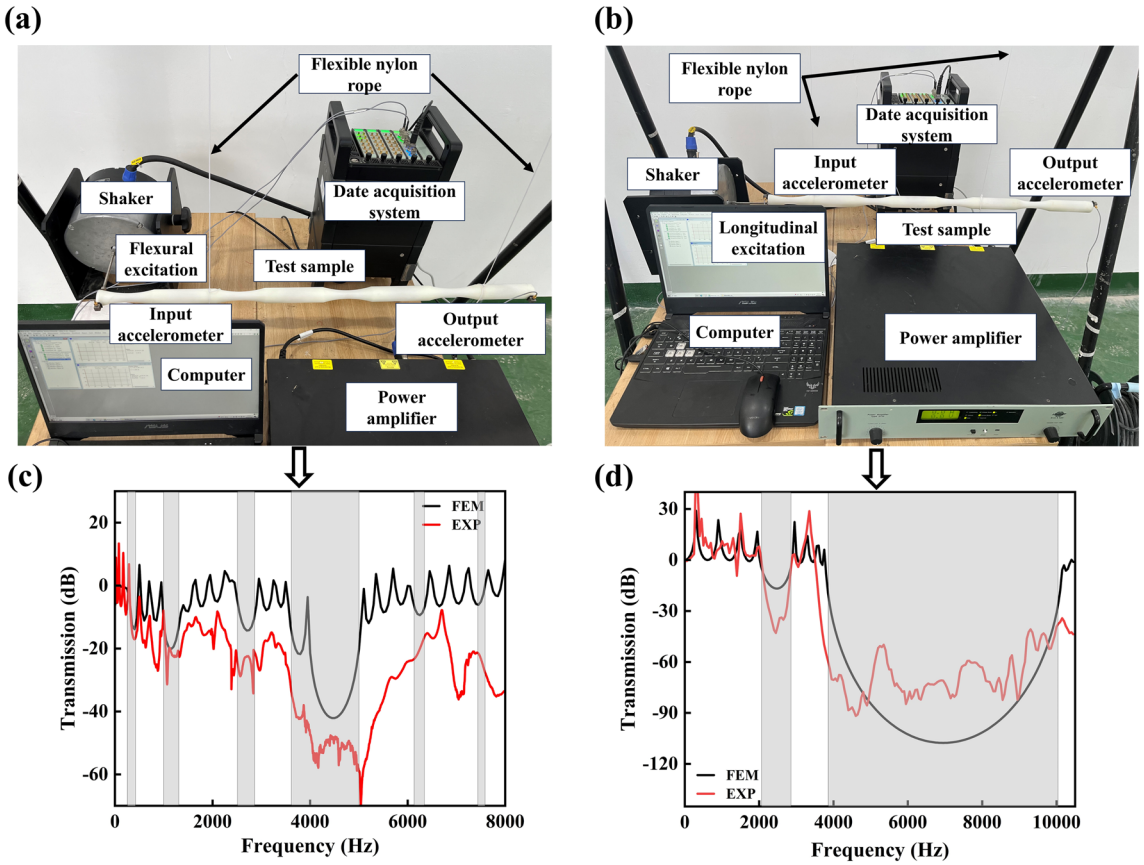


Fig. 13 **a** Photographs of the experimental process of the flexural vibration transfer experiment; **b** photographs of the experimental process of the longitudinal vibration transfer experiment; **c** comparison of the numerical results with the experimental results in the flexural direction (y direction) concerning the transmission loss; **d** comparison of the numerical results with the experimental results in the longitudinal direction (x direction) concerning the transmission loss

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curation, Methodology. Jingsheng Wang: Writing-review & editing, Resources. Xi Chen: Project administration. Kaiwen Chen: Investigation. Weirong Zhu: Resources.

Funding Major Program (JD) of Hubei Province, 2023BAA028, Shuang Yang.

Data Availability Data will be made available on request.

Declarations

Conflict of Interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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